

Mann §3.2 – §3.5

Lie Groups, Algebras, $SO(3)$, and the Appendix

A Slow, Heavily Annotated Walkthrough

Reading companion — Mann, *An Introduction to Particle Physics and the Standard Model*, Ch. 3

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Before you start: the big picture

Chapter 3 has been climbing a ladder:

- §3.1.1 — what a *group* is (abstract symbols + a rule).
- §3.1.2 — a *representation* turns those symbols into matrices.
- §3.1.3 — *irreducible* reps are the atomic building blocks.
- §3.1.4 — finite groups can be written as a multiplication table.

Now the chapter pivots. Most symmetries in physics — rotations, phase rotations, Lorentz boosts — have *infinitely many* elements. You can't write down a multiplication table with an uncountable number of rows. So §3.2–§3.5 build the machinery that handles continuous symmetries:

1. **§3.2 Lie groups** — smooth, continuous groups (rotations, etc.) and the idea that they are completely described by a small finite set of *generators*.
2. **§3.3 Algebras** — a more relaxed structure than a group: just a vector space with a way to combine vectors.
3. **§3.3.1 Lie algebras** — the algebra you get by zooming in on a Lie group at the identity. This is the central computational object.
4. **§3.4 $SO(3)$** — one full worked example: rotations in 3D, end to end.
5. **§3.5 Appendix** — the proof that closure of the Lie group $\Rightarrow$ the commutator relation of the Lie algebra.

The payoff: *instead of studying an infinite curved manifold of group elements, you only have to study a finite-dimensional vector space and the commutators of its basis vectors.* That replacement is what makes Lie theory tractable, and it is the central technical idea of the rest of the book.

1 §3.2 — Lie Groups

What's new compared to finite groups

A *finite* group has a discrete list of elements: $\{e, g_1, g_2, \dots, g_n\}$. A *Lie group* is a group whose elements depend *smoothly* on some continuous knobs θ^a .

Two pictures to keep in your head

- **Discrete (finite) group** – imagine a row of separated beads. You can only sit on a bead. Going from one bead to another is a discrete jump.
- **Lie group** – imagine a smooth curve, surface, or higher-dimensional sheet (a “manifold”). Every point on the sheet is a group element, and you can slide continuously between them. Group multiplication is a smooth function of where you are on the sheet.

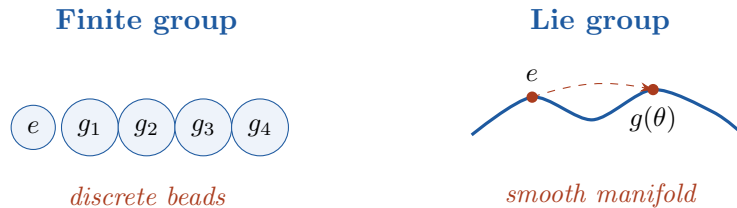


Figure 1: Finite group vs. Lie group. In the Lie group every point on the smooth curve is a separate group element; the parameter θ tells you where you are.

Mann’s defining formula, decoded slowly

Mann writes a typical Lie group element as

$$g = g(\theta^1, \dots, \theta^N) = \exp[i\theta^a T_a] = \exp[i\vec{\theta} \cdot \vec{T}]. \quad (3.1)$$

This formula is dense. Let’s take it one symbol at a time.

Reading equation (3.1) piece by piece

$\theta^1, \dots, \theta^N$

are continuous real *parameters* — the “knobs” you can turn. The number N is fixed by the group (for $SO(3)$, $N = 3$, one knob per rotation axis).

T_a , $a = 1, \dots, N$

are fixed matrices (or operators) called **generators**. There are exactly N of them, one per knob.

$\theta^a T_a$

uses the Einstein summation convention: $\theta^a T_a = \theta^1 T_1 + \theta^2 T_2 + \dots + \theta^N T_N$. Repeated indices are summed.

i

is the imaginary unit. The i is a physics convention that makes the T_a Hermitian (so they correspond to observables, e.g. angular-momentum components).

$\exp[\cdot]$

is the **matrix exponential**, defined by its power series:

$$\exp[M] = I + M + \frac{1}{2!}M^2 + \frac{1}{3!}M^3 + \dots$$

For an ordinary number this is the usual exponential; for a matrix it produces another matrix.

Why this is amazing

A continuous group has *infinitely many* elements (every value of every θ^a gives a different one). But equation (3.1) says all of them can be built out of a small fixed list of N matrices, the T_a . Know the N generators, know the whole group.

Compare:

- Finite group of order n : n entries in a multiplication table.
- Lie group: *infinitely many* elements, but only N generators (often $N \leq 10$) needed to encode them.

That's an enormous compression of information.

What “generator” means geometrically

Think of the Lie group as a curved surface (manifold). The identity element e corresponds to the origin (all $\theta^a = 0$). The generators T_a are like *tangent vectors at the identity*: they tell you which way you start moving when you turn the a^{th} knob slightly.

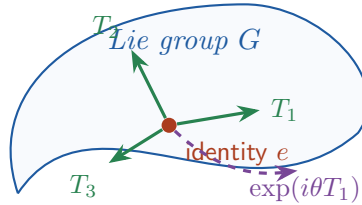


Figure 2: Geometrical picture. The generators T_a are tangent vectors at the identity. The exponential map $\exp(i\theta^a T_a)$ takes you *along* the manifold, tracing out an actual group element. The curved arrow shows the curve generated by turning only the T_1 knob.

Footnote (††): a subtle caveat

Mann’s footnote: the generators contain all the *local* information about the group, i.e. they reconstruct every element *continuously connected to the identity*. Some groups have “disconnected pieces” (e.g. the orthogonal group $O(3)$ contains rotations *and* reflections; you can’t continuously deform a rotation into a reflection). The generators only describe the piece containing the identity. The disconnected pieces matter in certain QED contexts (vacuum structure; ch. 25) but not for the bulk of particle physics.

Why Lie groups dominate physics

Most physical symmetries are *continuous*: you can rotate by any real angle, boost by any real velocity, multiply a wavefunction by any real phase. Mann gives the high-level taxonomy:

- **Spacetime symmetries** act on the spacetime coordinates (t, x, y, z) . Examples: rotations $SO(3)$, Lorentz boosts $SO(3, 1)$.
- **Internal symmetries** act on wavefunction components or on charges, leaving coordinates alone. Examples: $U(1)$ phase rotation in electromagnetism, $SU(3)$ color in QCD, $SU(2)$ weak isospin.

A streamlined version of Mann’s Table 3.4 (the catalog of Lie groups you will meet again and again in this book):

Group	Defining property	Where it shows up
$U(n)$	$n \times n$ unitary ($U^\dagger U = I$)	$U(1)$: electromagnetism
$SU(n)$	$U(n)$ with $\det U = 1$	$SU(3)$: strong, $SU(2)$: weak
$O(n)$	$n \times n$ orthogonal ($O^T O = I$)	$O(3)$: rotations + reflections
$SO(n)$	$O(n)$ with $\det O = 1$	$SO(3)$: rotations, $SO(3, 1)$: Lorentz
$Sp(n)$	$n \times n$ symplectic	phase-space structure (Hamilton’s mech)

Table 1: The standard Lie groups that organize particle physics. The “ S ” in SO / SU stands for “special” and means $\det = 1$.

Etymology cheat-sheet

- U = **Unitary** (preserves complex inner product; equivalent to: preserves probability).
- O = **Orthogonal** (preserves real inner product; equivalent to: preserves lengths and angles).
- S = **Special** (the additional restriction $\det = +1$; rules out reflections / orientation-flips).
- Sp = **Symplectic** (preserves a specific antisymmetric form; the language of Hamiltonian mechanics).

2 §3.3 — Algebras

What an algebra is (and isn’t)

An **algebra** is a more *relaxed* structure than a group:

- Start with a vector space V (so addition and scalar multiplication work the usual way).
- Add a binary “combining” operation $\circ$ that takes two vectors and produces a third *still in* V .

That’s it. **An algebra does not require an identity element. It does not require inverses.** It only requires *closure* (the result stays in the space).

$$v_I \circ v_J = \sum_K C^{IJ}_K v_K = C^{IJ}_K v_K \quad (3.2)$$

What equation (3.2) is really saying

Pick any two basis vectors v_I and v_J . Combine them with $\circ$. The result, being still in the vector space V , can be written as a linear combination of the basis vectors v_K . The coefficients of that combination, C^{IJ}_K , are called the **structure constants**. They are just numbers, and they tell you *everything* about how the algebra combines. Know the structure constants, know the algebra.

Concrete example: the vector cross product

Take V = vectors in 3D, with basis $\{\hat{x}, \hat{y}, \hat{z}\}$, and combining operation $\circ = \times$ (the cross product). Then

$$\hat{x} \times \hat{y} = \hat{z}, \quad \hat{y} \times \hat{z} = \hat{x}, \quad \hat{z} \times \hat{x} = \hat{y},$$

with all other cross products either zero or sign-flipped. This gives the structure constants $C^{IJ}_K = \epsilon_{IJK}$, the Levi-Civita symbol. So *3D vector-cross-product space is an algebra* — and a particularly important one, because we'll see shortly that it's the same algebra as the rotation group's.

Group vs. algebra in one line

Group:

closure + associativity + identity + inverses.

Algebra:

vector space + closure under $\circ$.

An algebra is what you keep when you throw away identities, inverses, and associativity-as-a-requirement, and keep just the rule for combining two things into a third. It is a *leaner* structure than a group.

§3.3.1 — Lie Algebras

The bridge: zooming in on a Lie group

Here is the central move of the whole chapter.

The key idea

A Lie group is a curved manifold. If you stand at the identity and *zoom in close enough*, the manifold looks flat — like a vector space. That tangent vector space, equipped with the right combining operation, is a **Lie algebra**. The Lie algebra is, in effect, the **linearization** of the Lie group near its identity.

Why bother? Because vector spaces are radically easier to work with than curved manifolds. By passing to the Lie algebra you replace nonlinear multiplication of group elements with *linear* algebra on a finite set of basis vectors.

The basis vectors of the Lie algebra turn out to be exactly the generators T_a we already met in equation (3.1). The combining operation is the **commutator**:

$$[T^a, T^b] \equiv T^a T^b - T^b T^a = if^{abc} T^c \quad (3.3)$$

Reading this slowly:

- $[T^a, T^b]$ is the *commutator* of two generators. It measures the failure of T^a and T^b to commute under multiplication.
- The right-hand side $if^{abc}T^c$ says: *the commutator is itself a linear combination of generators*. So the algebra is closed under the commutator (which is what makes it an algebra in the sense of §3.3).
- f^{abc} are the **structure constants** of the Lie algebra (the C^{IJ}_K of the previous section, specialized to this context). They are just numbers — but they completely encode the local geometry of the Lie group.

Jacobi identity

The associativity of group multiplication translates into a constraint on commutators:

$$[[T^a, T^b], T^c] + [[T^b, T^c], T^a] + [[T^c, T^a], T^b] = 0 \quad (3.4)$$

This is the **Jacobi identity**. Any vector space equipped with a bracket $[\cdot, \cdot]$ that is antisymmetric *and* obeys (3.4) is called a Lie algebra. The Jacobi identity is the algebra-level remnant of group associativity.

The whole point of Lie algebras, in two sentences

A Lie group's curved geometry is hard. Its Lie algebra is just a finite-dimensional vector space with a bracket, which is linear algebra — and linear algebra is something we can compute with. *Almost every calculation in particle physics symmetry is done at the Lie algebra level, not the Lie group level.*

3 §3.4 — Worked Example: the Rotation Group $SO(3)$

Now we do all of the above for one specific group, end to end. This is the most important worked example in the chapter — if you understand this section, you understand the rest.

Step 1: write down rotations as matrices

A rotation by angle θ about the $\hat{z}$ -axis sends

$$x' = \cos \theta x + \sin \theta y, \quad y' = -\sin \theta x + \cos \theta y, \quad z' = z.$$

In matrix form,

$$R_z(\theta) = \begin{pmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Similarly $R_x(\theta_x)$ and $R_y(\theta_y)$ are defined by rotations about $\hat{x}$ and $\hat{y}$. The full set of 3D rotations is what you get by multiplying these together with arbitrary $\theta_x, \theta_y, \theta_z$. This set is the group $SO(3)$ — 3×3 orthogonal matrices with $\det = +1$.

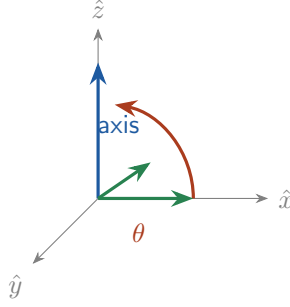


Figure 3: A rotation about the $\hat{z}$ -axis through angle θ . The parameter θ can be any real number; for each value you get a different element of $SO(3)$.

Step 2: zoom in near the identity to extract a generator

The identity element of $SO(3)$ is no rotation: $\theta = 0$. Near $\theta = 0$ we Taylor-expand $\cos \theta \approx 1$, $\sin \theta \approx \theta$:

$$R_z(\theta) \approx \begin{pmatrix} 1 & \theta & 0 \\ -\theta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I + \theta \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Mann wants the convention $R_z \approx I + i\theta T_z$, so we read off

$$T_z = -i \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (1)$$

That extra $-i$ is bookkeeping: it makes T_z *Hermitian*, so it deserves to be called a physical generator (its eigenvalues are real, just like an observable).

Repeating the same exercise for R_x and R_y gives the three generators of $SO(3)$:

$$T_x = -i \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad T_y = -i \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T_z = -i \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

These three matrices are a *basis* for the Lie algebra $\mathfrak{so}(3)$.

Why three?

The dimension of the Lie algebra equals the number of independent parameters of the Lie group. $SO(3)$ has three (one rotation angle per axis: $\theta_x, \theta_y, \theta_z$). So $\mathfrak{so}(3)$ has three basis vectors. In general, an N -parameter Lie group has an N -dimensional Lie algebra.

Step 3: compute a commutator (the hard step, fully shown)

We want to verify $[T_x, T_y] = iT_z$. Just plug in. With $(-i)^2 = -1$:

$$T_x T_y = (-i)^2 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = - \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$T_y T_x = (-i)^2 \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} = - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Subtracting:

$$[T_x, T_y] = T_x T_y - T_y T_x = - \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

And $iT_z = i \cdot (-i) \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. They match. So $[T_x, T_y] = iT_z$.

By symmetry the other two commutators come out $[T_y, T_z] = iT_x$ and $[T_z, T_x] = iT_y$. All three can be packaged into the single formula

$$[T^a, T^b] = i \epsilon^{abc} T^c \quad (3.10)$$

where the structure constants $f^{abc} = \epsilon^{abc}$ are exactly the Levi-Civita symbol from the cross-product example earlier in §3.3. **The Lie algebra $\mathfrak{so}(3)$ has the same structure constants as the 3D vector cross product.** That is not a coincidence: infinitesimal rotations *are* cross products with the rotation axis.

Step 4: convention — rename $T_a \rightarrow J_a$

The algebra $\mathfrak{so}(3)$ and the group $SO(3)$ are so ubiquitous in physics — they are the angular momentum algebra you saw in quantum mechanics — that the generators get a special name:

$$T_a \longrightarrow J_a, \quad [J^a, J^b] = i \epsilon^{abc} J^c. \quad (3.13)$$

These are the angular momentum operators. The hydrogen-atom commutation relations $[J_x, J_y] = i\hbar J_z$ (with $\hbar = 1$ in natural units) are literally this equation.

Step 5: exponentiate back to recover a finite rotation

The exponential map walks you from the algebra back to the group. With $\vec{\theta} = \theta \hat{y}$:

$$\exp(i\theta J_y) = I + (i\theta J_y) + \frac{1}{2!}(i\theta J_y)^2 + \cdots$$

Mann grinds through the series; the trigonometric pattern emerges because J_y^2 acts as a projector that eats only the x, z subspace. The final result is

$$\exp(i\theta J_y) = \begin{pmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{pmatrix}. \quad (3.17)$$

This is exactly a rotation by θ about the $\hat{y}$ -axis — exactly the kind of matrix we started with. So the round trip

$$\text{rotation matrix} \rightarrow \text{generator } J_y \rightarrow \text{rotation matrix}$$

closes up. The exponential map is the bridge from the algebra back to the group.

The structural summary of §3.4

1. **Start with the group:** matrices R_x, R_y, R_z .
2. **Linearize at the identity:** extract generators J_x, J_y, J_z .
3. **Compute commutators:** $[J^a, J^b] = i\epsilon^{abc} J^c$.
4. **Exponentiate:** $\exp(i\theta^a J_a)$ reproduces any rotation.

This is the playbook you'll use again for $SU(2)$, $SU(3)$, the Lorentz group, etc.

4 §3.5 — Appendix: Why the Commutator Relation Holds

This appendix proves that the commutator relation (3.3) is a *consequence* of the Lie group being closed under multiplication. You don't need this for computations, but it explains *where the brackets come from*.

Setup: closure of the Lie group

Pick two group elements $g_1 = \exp(i\Theta^a T_a)$ and $g_2 = \exp(i\Upsilon^b T_b)$. Their product is still in the group, so it must itself be expressible as $\exp$ of something built from the generators:

$$\exp(i\Theta^a T_a) \exp(i\Upsilon^b T_b) = \exp(i\Xi^c(\Theta, \Upsilon) T_c) \quad (3.20)$$

where $\Xi^c(\Theta, \Upsilon)$ is some function of the two sets of parameters.

Two sanity checks on Ξ

- If $\Upsilon = 0$ then the second factor is the identity, so $\Xi^c(\Theta, 0) = \Theta^c$.
- If $\Theta = 0$ similarly $\Xi^c(0, \Upsilon) = \Upsilon^c$.

So at lowest order $\Xi^c = \Theta^c + \Upsilon^c$. What's the first correction? Mann argues from parity considerations that Ξ^c can only contain odd-power cross-terms in Θ and Υ . The leading correction is therefore quadratic:

$$\Xi^c(\Theta, \Upsilon) = \Theta^c + \Upsilon^c - \frac{1}{2} f^{abc} \Theta^a \Upsilon^b + \dots \quad (3.21)$$

The coefficients f^{abc} are just *names* for those unknown expansion numbers — we'll see they turn out to be exactly the structure constants.

Match terms on both sides

Expand both sides of (3.20) to second order in Θ and Υ . The expansion of the left side gives (collecting only the cross term $\Theta^a \Upsilon^b$):

$$-\Theta^a \Upsilon^b T_a T_b.$$

The expansion of the right side gives the cross term

$$-\frac{1}{2}\Theta^a\Upsilon^b(T_aT_b + T_bT_a) - \frac{i}{2}f^{abc}\Theta^a\Upsilon^bT_c.$$

Setting the two equal and cancelling the symmetric piece gives

$$-\frac{1}{2}\Theta^a\Upsilon^b(T_aT_b - T_bT_a) = -\frac{i}{2}f^{abc}\Theta^a\Upsilon^bT_c.$$

Since this must hold for every Θ and Υ , the coefficients must match:

$$[T^a, T^b] = if^{abc}T^c. \quad (3.24)$$

And that is the commutation relation (3.3) we postulated earlier.

What the appendix actually proves

Closure of the Lie group $\Rightarrow$ commutator structure of its Lie algebra. The mysterious-looking equation (3.3) isn't an extra axiom we have to demand — it falls out automatically the moment we insist that the product of two group elements is itself a group element and expand to second order. The structure constants f^{abc} are the *quadratic-order correction* to how parameters combine when you multiply.

The headline of the entire chapter

A Lie group encodes its full local structure in a finite set of generators T_a and their commutators $[T^a, T^b] = if^{abc}T^c$. That data set — a small vector space plus a bracket — is the Lie algebra. From the Lie algebra you can recover any Lie group element via the exponential map $\exp(i\theta^aT_a)$. So in the rest of the book, whenever Mann writes “the symmetry group is G ,” you should think: *this means a vector space of generators, with a commutator table of structure constants, and an exponential map back to the group.*

Glossary at a glance

Term	Plain-English meaning
Lie group G	a group whose elements depend smoothly on continuous parameters θ^a
Parameter θ^a	one of the continuous knobs that selects a group element
Generator T_a	a fixed matrix; tangent vector to the group manifold at the identity
Exponential map	$g(\theta) = \exp(i\theta^a T_a)$; the bridge from algebra to group
Algebra	vector space + closed binary operation; weaker than a group (no identity, no inverses required)
Structure constants f^{abc}	numbers that encode the commutator $[T^a, T^b] = i f^{abc} T^c$
Lie algebra	vector space of generators + commutator bracket; the “linearization” of a Lie group at the identity
Jacobi identity	$[[T^a, T^b], T^c] + [[T^b, T^c], T^a] + [[T^c, T^a], T^b] = 0$; the algebra-level remnant of group associativity
$\mathfrak{so}(3)$	Lie algebra of $SO(3)$; generators J_a with $[J^a, J^b] = i\epsilon^{abc} J^c$; same as quantum angular momentum